

CARLOS MIGUEL FERNÁNDEZ GARCÍA

Email: carlos.fernandez@gmail.com

Phone: +34 689 447 238

LinkedIn: linkedin.com/in/carlos-fernandez-garcia-barcelona

GitHub: github.com/cmfgarcia

PROFESSIONAL SUMMARY

I'm a research engineer based in Barcelona with a passion for solving complex technical problems through experimentation and data analysis. I enjoy working at the intersection of machine learning and systems optimization, where I can contribute to projects that have measurable impact. My background spans cloud infrastructure, distributed systems, and applied research, and I'm most motivated when collaborating with teams to turn experimental ideas into production solutions.

WORK EXPERIENCE

Research Engineer - Advanced Systems Lab

TechVision Research, Barcelona, Spain

www.techvisionresearch.io

January 2022 - Present

Designed and implemented a distributed machine learning pipeline processing 500GB of daily sensor data, reducing inference latency by 40% through optimized model quantization and edge deployment strategies.

Conducted research on federated learning approaches for privacy-preserving data analysis, resulting in two peer-reviewed publications at the ECML 2023 conference.

Led cross-functional collaboration with the backend team to integrate PyTorch models into a microservices architecture, establishing MLOps best practices and reducing model deployment time from 3 weeks to 3 days.

Mentored two junior engineers on research methodologies and helped them develop their first published research paper on anomaly detection in time-series data.

Senior Software Engineer - Cloud Infrastructure Team

Nexus Cloud Solutions

nexuscloud.com

August 2020 - December 2021

Architected and deployed a Kubernetes-based container orchestration platform serving 200+ internal microservices with 99.99% uptime, handling peak loads of 50,000 requests per second.

Optimized CI/CD pipelines using GitLab CI and Terraform, reducing deployment time by 55% and enabling 50+ releases per day across all development teams.

Implemented comprehensive monitoring and observability solutions using Prometheus, Grafana, and ELK stack, decreasing incident response time from 45 minutes to 8 minutes average.

Authored internal documentation and conducted training sessions on containerization best practices, establishing standards adopted across engineering organization of 150+ engineers.

Junior Research Engineer - Data Science Division

InnovateLabs Europe

innovatelabs.net

March 2019 - July 2020

Developed Python-based data processing pipelines using Apache Spark to clean and prepare datasets containing 2 billion records for machine learning experiments, achieving 99.8% data quality scores.

Implemented computer vision models using TensorFlow for real-time object detection in video streams, achieving 94% accuracy and processing 30 frames per second on GPU infrastructure.

Collaborated on research initiatives exploring transfer learning techniques, experimenting with 15+ pre-trained models and publishing benchmark results in internal technical blog reaching 500+ monthly readers.

Supported infrastructure team in managing AWS resources including EC2, S3, and RDS, ensuring cost optimization while maintaining performance requirements for experimental workloads.

EDUCATION

Master of Science in Computer Science, Autonomous University of Barcelona (UAB)

Specialization: Machine Learning and Data Science

Graduated: June 2019

Thesis: Federated Learning Approaches for Distributed Data Analysis

Bachelor of Science in Information Technology, Universitat Politècnica de Catalunya (UPC)

Graduated: May 2017

TECHNICAL SKILLS

Programming Languages: Python, Java, Go, SQL, Bash

Machine Learning Frameworks: PyTorch, TensorFlow, scikit-learn, XGBoost

Cloud and Infrastructure: AWS (EC2, S3, Lambda, RDS), Google Cloud Platform, Docker, Kubernetes, Terraform

Data Processing: Apache Spark, Pandas, NumPy, PostgreSQL, MongoDB

DevOps and Tools: GitLab CI, Jenkins, Prometheus, Grafana, ELK Stack

Research and Analysis: Jupyter Notebook, Matplotlib, Seaborn, Statistical Analysis, Experimental Design

GITHUB REPOSITORIES

cmfgarcia/federated-learning-framework - Implementation of federated learning protocols with privacy differential privacy guarantees, 342 stars, 89 forks

cmfgarcia/edge-ml-optimizer - Lightweight framework for optimizing neural networks for edge device deployment, 156 stars, 41 forks

cmfgarcia/time-series-anomaly-detection - End-to-end solution for detecting anomalies in multivariate time-series data using ensemble methods, 98 stars, 27 forks

cmfgarcia/distributed-data-pipeline - Apache Spark-based ETL pipeline for processing and validating large-scale datasets with data quality metrics, 67 stars, 18 forks

CERTIFICATIONS AND AWARDS

AWS Certified Solutions Architect Associate, 2021

Google Cloud Professional Data Engineer, 2022

Best Research Poster Award, InnovateLabs Annual Conference, 2020